

TRABAJO:

$$W = \vec{F} \cdot \vec{x} \quad \text{Unidad (J)} \quad \text{Nm} \rightarrow \text{Julios} \quad \vec{A} \cdot \vec{B} \quad \text{Si} \quad \vec{A} = (A_x, A_y, A_z) \quad \vec{B} = (B_x, B_y, B_z) \Rightarrow \vec{A} \cdot \vec{B} = (A_x B_x + A_y B_y + A_z B_z)$$

Ejemplo 1:

$$W = (3, -2, 1) \cdot (2, 2, 2)$$

$$W = (3 \cdot 2) - (2 \cdot 2) + 2 = 6 - 4 + 2 = 4 \text{ (J)}$$

$$\vec{F} = (3, -2, 1) \text{ (N)} \Rightarrow |\vec{F}| = \sqrt{3^2 + (-2)^2 + 1^2} = \sqrt{9 + 4 + 1} = \sqrt{14} \text{ (N)}$$

$$\vec{x} = (2, 2, 2) \text{ (m)} \Rightarrow |\vec{x}| = \sqrt{(2)^2 + (2)^2 + (2)^2} = \sqrt{2^2(1+1+1)} = \sqrt{4 \cdot 3} = 2\sqrt{3} \text{ (m)}$$

$\Theta = ?$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \Theta$$

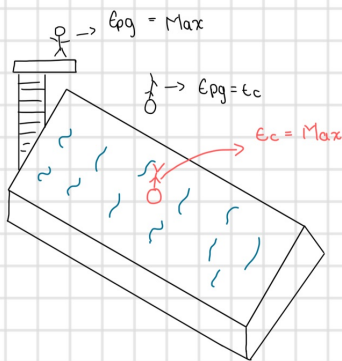
$$\frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \cos \Theta$$

$$\Theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} \right) = 72,024^\circ$$

CONSERVACIÓN de ENERGÍA:

Energía antes = Energía después

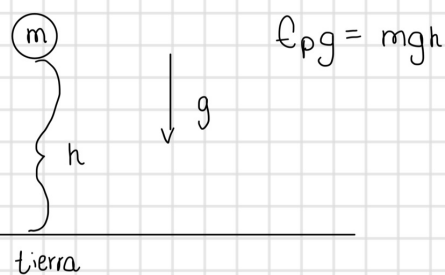
Ejemplo 1:



Conservación de energía:

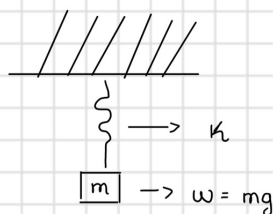
$$mgh = \frac{1}{2} m v_1^2 + mgh_2 \quad \cdot \quad gh_1 = \frac{1}{2} v_1^2 + gh_2$$

Energía potencial gravitatoria



Energía elástica:

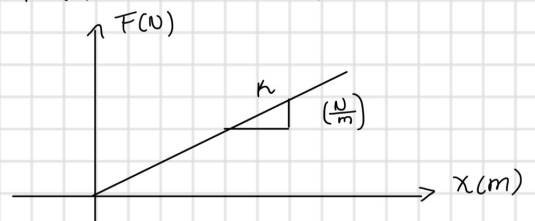
Ley de Hooke



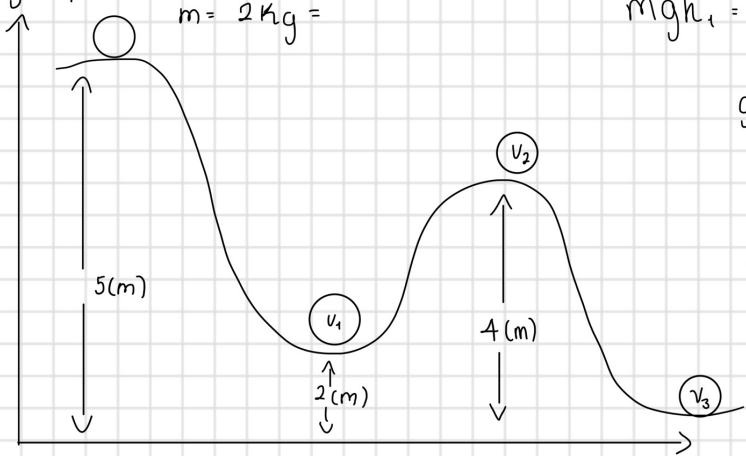
$$F_r = -Kx$$

$$Epe = \frac{1}{2} Kx^2$$

$x \text{ (m)}$	x_1	x_2	$\dots$	x_n
$F \text{ (N)}$	$m_1 g$	$m_2 g$	$\dots$	$m_n g$



Ejemplo 2: $m = 2 \text{ kg} =$



$$mgh_1 = \frac{1}{2}mv_1^2 + mgh_2$$

$$gh_1 = \frac{1}{2}v_1^2 + gh_2$$

$$\sqrt{2(gh_1 - gh_2)} = v_1$$

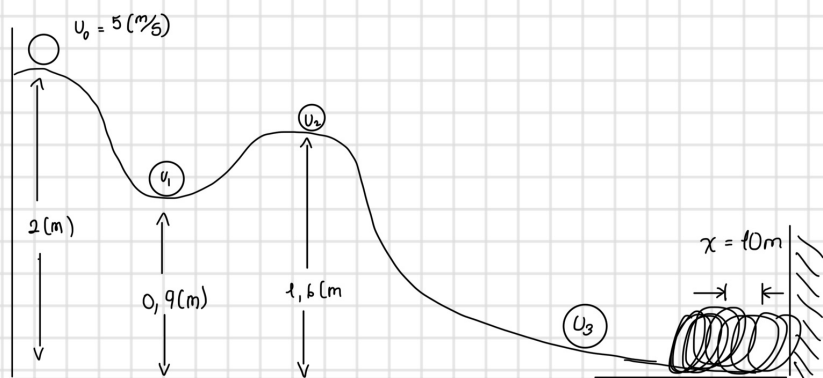
$$v_1 = \sqrt{2(9,8(5) - 9,8(2))} \approx 7,668 \text{ m/s}$$

$$E_c = \frac{1}{2}mv^2$$

$$E_{pg} = mgh$$

$$E_{pe} = \frac{1}{2}kx^2$$

Ejemplo 3:



Datos:

$$v_0 = 5 \text{ (m/s)} \quad x = 10 \text{ cm} =$$

$$h_1 = 2 \text{ (m)}$$

$$h_2 = 0,9 \text{ (m)}$$

$$h_3 = 1,6 \text{ (m)}$$

$$mgh_1 + \frac{1}{2}mv_0^2 = mgh_2 + \frac{1}{2}mv_1^2$$

$$gh_1 + \frac{1}{2}v_0^2 = gh_2 + \frac{1}{2}v_1^2$$

$$(9,8)(2) + \frac{1}{2}(5)^2 = (9,8)(0,9) + \frac{1}{2}v_1^2$$

$$v_1 = 6,823 \text{ (m/s)}$$

$$gh_2 + \frac{1}{2}v_1^2 = gh_3 + \frac{1}{2}v_2^2$$

$$(9,8)(0,9) + \frac{1}{2}(6,823)^2 = (9,8)(1,6) + \frac{1}{2}v_2^2$$

$$v_2 = 5,730 \text{ (m/s)}$$

$$gh_3 + \frac{1}{2}v_2^2 = \frac{1}{2}v_3^2$$

$$(9,8)(1,6) + \frac{1}{2}(5,730)^2 = \frac{1}{2}v_3^2$$

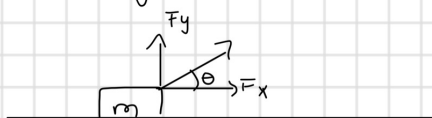
$$v_3 = 8,012 \text{ (m/s)}$$

$$mv_3^2 = kx^2 \rightarrow k = \frac{m(v_3)^2}{x^2}$$

$$k = \frac{(0,5)(8,012)^2}{(0,1)^2}$$

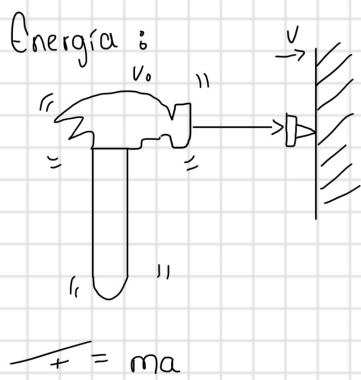
$$k = 3209,607 \text{ (N/m)}$$

trabajo:



$$W \rightarrow W_x = F_x \cdot x = F_x \cos \theta$$

$$W_y = F_y \cdot y = F_y \sin \theta = 0$$



$$W = F \cdot d$$

$$W = mad$$

$$W = m \left(\frac{-v_0^2}{2d} \right) d$$

$$W = m \left(\frac{-v_0^2}{2} \right)$$

$$v^2 = v_0^2 + 2ad$$

$$0 = v_0^2 + 2ad$$

$$-v_0^2 = 2ad$$

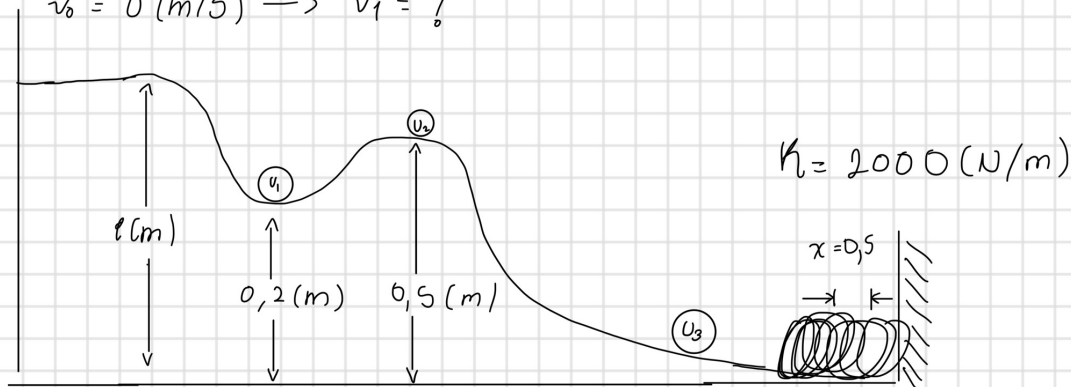
$$\frac{-v_0^2}{2d} = a$$

$K \rightarrow$ Cinética $\rightarrow E_c$

$$E_c = \frac{1}{2} mv^2$$

Ejemplo 3:

$$v_0 = 0 \text{ (m/s)} \rightarrow v_1 = ?$$



$$m = 0.5 \text{ (kg)}$$

$$m \cdot gh + \frac{1}{2} m v_1^2 = \frac{1}{2} K x^2$$

$$(0.5)(9.8) + \frac{1}{2} (0.5) v_1^2 = \frac{1}{2} (2000) (0.5)^2$$

$$v_1 = 31.31 \text{ (m/s)}$$